

Karary University

College of Computer science and Technology

Software Engineering Department

Third Level

هندسة وتصميم تطبيقات الإنترنت

INTERNET APPLICATIONS ENGINEERING AND DESIGN

Abobakr A Talha



Basics of Web Design

Chapter 1
Internet & Web Basics
Key Concepts

Lecture No (1)



Learning Outcomes

- ⦿ Describe the evolution of the Internet and the Web
- ⦿ Explain the need for web standards
- ⦿ Describe universal design
- ⦿ Identify benefits of accessible web design
- ⦿ Describe the purpose of web browsers and web servers
- ⦿ Identify Internet protocols
- ⦿ Define URIs and domain names
- ⦿ Describe XHTML and HTML
- ⦿ Create your first web page
- ⦿ Use the body, head, title and meta elements
- ⦿ Name, save, and test a web page

The Evolution of the Internet

- **Internet**
 - **Inter**connected **net**work of computer networks
 - **ARPAnet**
 - Advanced Research Project Agency
 - 1969 – four computers connected
 - **NSFnet**
 - National Science Foundation
 - Use of the Internet was originally **limited** to government, **research** and **academic** use
 - 1991 Commercial ban lifted



Reasons for Internet Growth in the 1990s

- Removal of the ban on commercial activity
- Development of the **World Wide Web** by Tim Berners-Lee at CERN
- Development of Mosaic, the first graphics-based web browser at NCSA
- Convergence of technologies:
 - Affordable personal computers with GUI Operating Systems
 - Affordable Internet service providers

Web Standards and the W3C Consortium



W3C – **W**orld **W**ide **W**eb **C**onsortium

- Develops recommendations and prototype technologies related to the Web
- Produces specifications, called Recommendations, in an effort to **standardize web technologies**
- **WAI** – **W**eb **A**ccessibility **I**nitiative



Web Accessibility

“The power of the Web is in its universality. Access by everyone regardless of disability is an essential aspect.”
– Tim Berners-Lee

- ④ **Accessible Websites**

provide accommodations that help individuals to individuals with visual, auditory, physical, and neurological disabilities overcome barriers

Network Overview



Network

two or more computers connected together for the purpose of communicating and sharing resources

Internet, Packets and Routing

- **Internet** is a network of computer networks
- Data is transmitted by packet switching using the standard **Internet Protocol (IP)**
- **Packet** – a unit of information carriage
- **Packet switching** – process of moving packets from one node (computer device) to another

Internet, Packets and Routing

- At the sender, data is **broken into packets** and sent to the nearest node (**router**)
- At each router, it sends the packet to another router that is **closer** to the **final destination**
- At the receiver, packets are **reassembled** to get the **original data**
- A simple analogy: mailing system

Internet v.s. Web

- **The Internet**

a inter-connected computer networks, linked by **wires**, **cables**, **wireless connections**, etc.

- **Web**

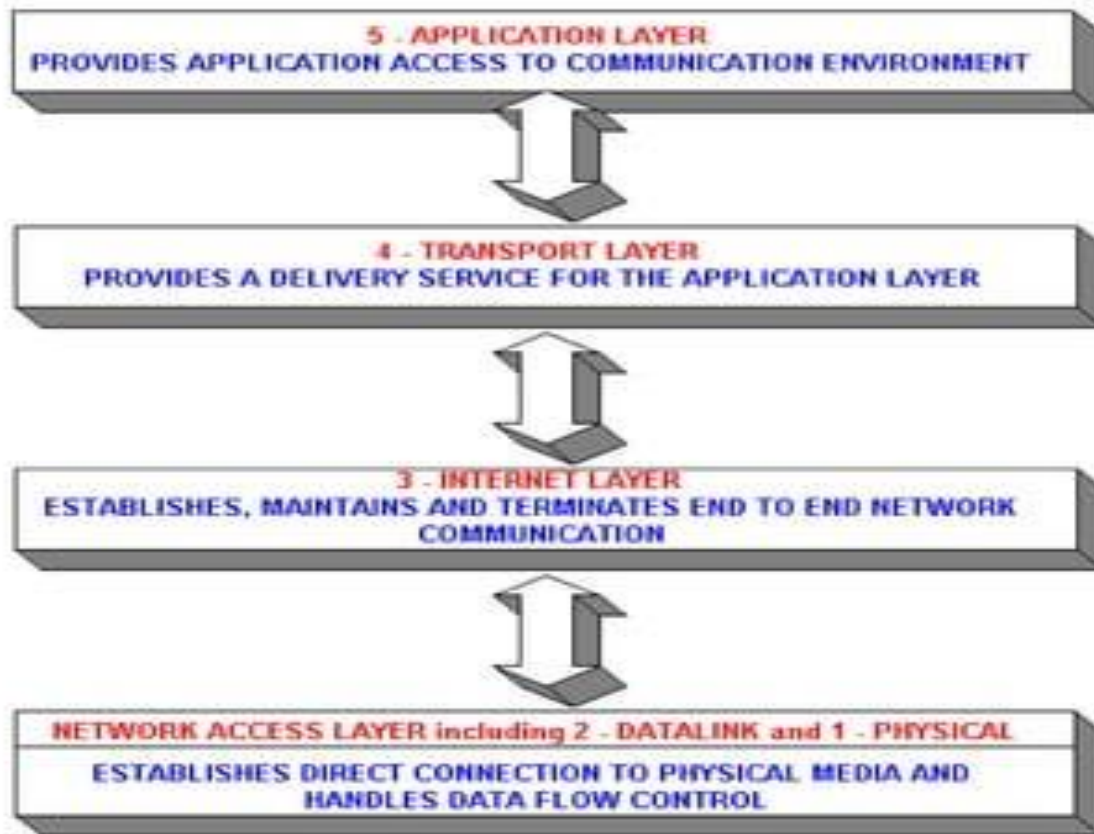
a collection of **interconnected documents** and other resources.

- The world wide web (**WWW**) is accessible via the Internet, as are many **other services** including **email**, **file sharing**, etc.

How does the Internet Work?

- Through
- **communication protocols**
 - set of Rules** that describe the methods used for clients and servers to **communicate** with each other over a network.
 - **IP** (Internet Protocol)
 - **TCP** (Transmission Control Protocol)
 - **UDP** (User Datagram Protocol)
 - **DNS** (Domain Name Service),
 - **SMTP** (Simple Mail Transmission Protocol)
 - **FTP** (File Transmission Protocol)

TCP/IP Protocol Suites



HTTP, FTP, Telnet, DNS, SMTP, etc.

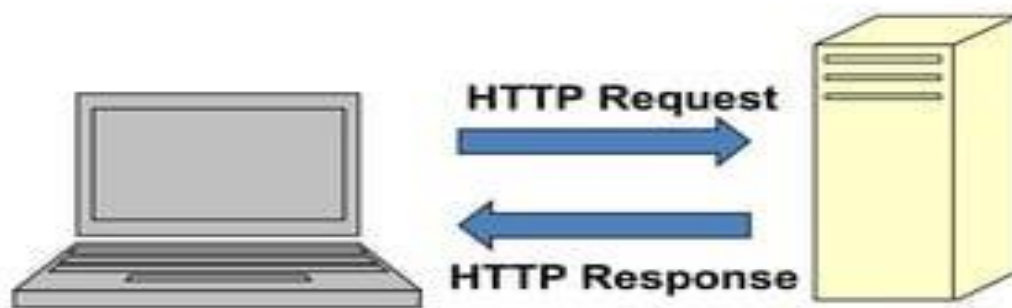
TCP, UDP

IP (IPv4, IPv6)

HTTP

Hypertext Transfer Protocol

- A set of rules for exchanging files such as text, graphic images, sound, video, and other multimedia files on the Web.



- Web browsers send HTTP requests for web pages and their associated files.
- Web servers send HTTP responses back to the web browsers.

IP Address

- Each device connected to the Internet has a **unique numeric IP address**.
- These addresses consist of a set of **four groups of numbers**, called octets.
74.125.73.106 will get you Google!
- An **IP address** may **correspond** to a **domain name**.

The World Wide Web (WWW)

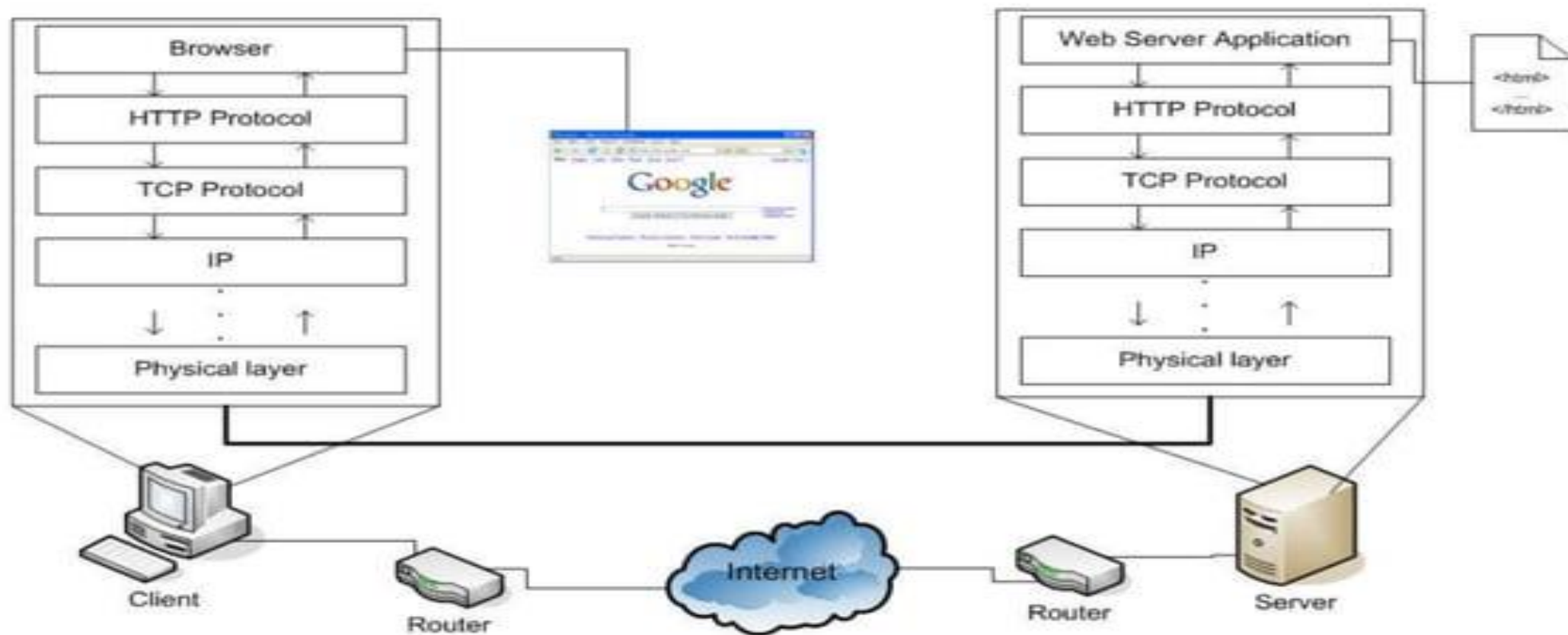


- The **World Wide Web** (commonly shortened to the **Web**) is a system of interlinked, hypertext documents accessed via the Internet.
- It is created to **share files/documents** and **overcome** the **barrier of different file formats**
- **Hypertext** refers to text on a computer that will lead the user to other, related information on demand.

Basics of the WWW

- **Hypertext:**
a format of information which allows one to move from one part of a document to another or from one document to another through **hyperlinks**
- **Uniform Resource Locator (URL):**
unique identifiers used to locate a particular resource on the network
- **Markup language:**
 - defines the structure and content of hypertext documents. Ex: HTML
- **A browser**
is a **software program** which **interprets** the **HTML documents** and **displays** it on the user's screen.

Putting it All Together



Client-Server Model

- WWW use classical client / server architecture
- **Client** :web browsers, used to surf the Web
- **Server** :web server , used to supply information to browsers

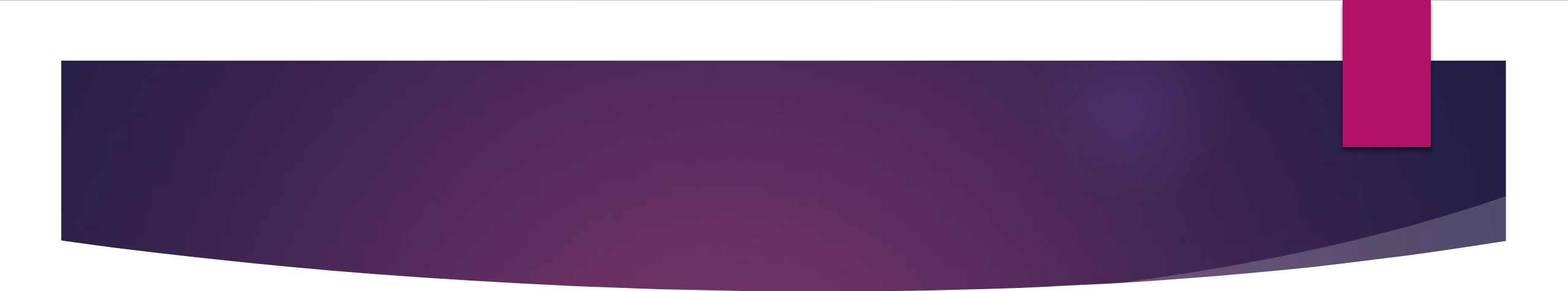


How do Web Servers work?

- Client specifies document at a specific web address that is desired (specified by a URL)
 - Ex: <http://www.cs.pitt.edu/>
- If the document is HTML or text, the server simply forwards it back to the client
 - If it is **text**, it is shown unaltered in the browser
 - If it is **HTML** it is **rendered** in the client's browser
 - HTML tags are interpreted and result is shown to the user

What is a Web Page?

- **Web pages** are text files containing **HTML**
- **HTML** – Hyper Text Markup Language
 - A notation for describing
 - **document structure** (semantic markup)
 - **formatting** (presentation markup)
- The **markup tags**
provide information about the **page content structure**



The End